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1. Cover Page

Field	Detail
Project Codename	DeliverIQ
Project Title	Severity Reality Check
Subtitle	Disentangling True Accident Severity from Congestion Bias
Sector	Transportation & Public Safety
Section	A
Team ID	G12
Faculty Mentor	*Fill mentor name*
Institute	Newton School of Technology
GitHub Repository	https://github.com/shiavm006/Section-A_G12_DeliverIQ
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2. Executive Summary

Public road-accident datasets are widely used by insurers, logistics planners, and state Departments of Transportation (DOTs) to assess corridor risk, set premiums, and allocate infrastructure dollars. The most-used severity field in the largest US accidents dataset, however, measures *traffic-flow impact* (short delay vs. long delay) – **not** physical crash severity. A fender-bender at rush hour can register as Severity 4, while a fatal head-on collision on a quiet rural road

may register as Severity 1. Decisions taken on the raw signal therefore systematically over-weight congested urban corridors and under-invest in genuinely high-risk locations.

The project applies an operational definition of "True Severe" – a Severity-4 accident whose road impact extends at least half a mile – and quantifies how much of the conventional severity signal is actually congestion noise.

Key findings

- Roughly **half of all accidents flagged as Severity 4 fail the True-Severe threshold.** Public severity counts are inflated by congestion artifacts.
- **Average severity peaks at midnight (12 AM)**, not during evening rush hour. Volume peaks at 5 PM; *risk* peaks 6-7 hours later.
- **Pennsylvania, not California, leads** the True-Severe leaderboard. PA's per-accident severity is roughly 3 x the national average despite ~3 % of total accident volume.
- **Most severe accidents occur in clear weather** (avg visibility ~ 9.1 mi). Conventional wisdom that fog / storm conditions drive severity is largely wrong in this dataset.
- **Traffic signals reduce per-accident severity by ~0.4 points** – a quantifiable, ROI-positive infrastructure intervention.

Recommended actions

1. Insurers should down-weight publicly reported Severity 4 by ~50 % when used as a pricing input.
2. State DOTs should prioritize signal installations at top-N junction hotspots (Signal Drop ~ 0.4 severity points = measurable ROI).
3. Logistics fleets should re-rank corridors using a True-Severe Index, not raw counts.
4. Federal grant programs should reweight per-state allocations using True Severe rather than total volume.
5. Infrastructure planners should rebalance away from visibility-only interventions (where the effect is small) toward junction redesigns and signal coverage.

3. Sector & Business Context

The US road network supports more than **3 trillion vehicle-miles per year**, and the FHWA estimates traffic crashes cost the US economy more than **$1 trillion annually** when medical care, lost productivity, congestion, and property damage are included. The auto-insurance industry alone underwrites more than **$300 billion in annual premiums**, much of it priced using publicly available crash and incident data as one input.

Three stakeholder groups make daily decisions on this data:

- **Property-and-Casualty (P&C) insurers** ingest public severity feeds into pricing models. The accuracy of those feeds directly affects premium fairness, loss ratios, and adverse-selection risk.
- **State Departments of Transportation (DOTs)** allocate billions in capital each year to signals, signage, lighting, and lane redesigns. Severity-weighted hotspot maps are a primary input.
- **National logistics fleets and last-mile carriers** route around high-risk corridors and adjust dispatch windows. Mis-priced corridor risk leads to either over-cautious routing (lost time) or under-cautious routing (claim spikes).

Why this problem matters now. Two trends amplify the cost of acting on biased severity data: (1) accident-reporting APIs increasingly feed real-time pricing engines, so any systematic bias compounds quickly; and (2) the post-COVID rebound in vehicle-miles-traveled has pushed accident volume back to pre-pandemic levels, meaning more decisions, more premiums, and more capital flow through the same biased input.

The decision-maker this report serves is the **risk and pricing team at a national P&C insurer**, with secondary applicability to state DOT planners and national fleet operators. The corrected severity signal proposed here is portable across all three.

4. Problem Statement & Objectives

4.1 Formal Problem Definition

> **Public severity classifications in US accident data conflate traffic-flow impact with physical crash severity. Decisions made directly on the raw severity field – including premium pricing, infrastructure investment, and corridor risk ranking – are systematically biased toward congested urban locations and away from genuine crash hotspots. The objective of this project is to (a) quantify the magnitude of that bias, (b) produce a corrected severity signal that decision-makers can use, and (c) translate the findings into specific, actionable recommendations for insurers, DOTs, and logistics planners.**

4.2 Scope

In scope

- US Accidents (2016 - 2023) dataset, sampled to 95,607 rows balanced across all 4 severity tiers
- Disentangling severity from time-of-day, day-of-week, season, and rush-hour confounds
- Disentangling severity from geographic confounds (state, region, city)
- Disentangling severity from environmental confounds (weather, visibility, temperature)
- Disentangling severity from infrastructure confounds (junctions, traffic signals, crossings, etc.)
- Three stakeholder lenses: insurer / DOT / logistics

Out of scope

- Real-time prediction (this is a retrospective analysis, not a live model)
- Casualty / fatality outcomes (the dataset does not include injury counts)
- Causal inference beyond statistical association
- Cost-benefit modelling of specific interventions (we estimate impact directionally, not in dollar terms)

4.3 Success Criteria

The project succeeds if it can:

1. Quantify the share of nominal Severity 4 that fails a defensible "True Severe" threshold.
2. Identify at least one counter-intuitive finding (e.g., peak-risk hour != rush hour).
3. Produce a Tableau dashboard with at least one interactive filter that lets a stakeholder explore the corrected signal.
4. Translate the analytical output into 3 - 5 specific, decision-ready recommendations.

All four criteria are met (see Sections 11 - 13).

5. Data Description

5.1 Source

- **Dataset**: *US Accidents (2016 - 2023)* by Sobhan Moosavi
- **Direct access**: https://www.kaggle.com/datasets/sobhanmoosavi/us-accidents
- **License**: Creative Commons BY-NC-SA 4.0 (academic / research use only)

The dataset was assembled from multiple traffic-incident APIs (state DOT feeds, law-enforcement reporting systems, traffic cameras, and roadside sensors).

5.2 Structure

Attribute	Value
Original size	~7.7 million rows x 46 columns (3.06 GB CSV)
Working sample	95,607 rows x 49 columns (post-cleaning + feature engineering)
Time period	February 2016 - March 2023
Geographic coverage	49 contiguous US states; 6,800+ cities
Primary granularity	One row per reported traffic accident

5.3 Why a Sampled Dataset

The full 7.7 M-row file is highly imbalanced. Severity 2 represents ~80 % of all records; Severity 1 and Severity 4 are rare (<1 % and <3 % respectively). Statistical analysis on the raw distribution would be dominated by the majority class.

We extracted a **balanced stratified sample** in `notebooks/01_extraction.ipynb`:

Severity	Target rows	Rationale
1	10,000	Over-sample to retain rare class
2	35,000	Down-sample to control majority dominance
3	35,000	Match Severity 2 for cross-class comparison
4	20,000	Over-sample to support True Severe analysis

Within each severity stratum, sampling preserved geographic diversity (proportional representation across all 49 states).

5.4 Key Columns

Column	Type	Role
`Severity`	int	Target field; traffic-flow impact (1 = short delay, 4 = long delay)
`Distance_mi`	float	Length of road segment affected; defines `Is True Severe`
`Start_Time`	datetime	Source of `Year`, `Month`, `Hour`, `Day_of_Week`, `Is_Rush_Hour`
`State`, `City`, `County`	string	Geographic location; supports state / region / city analysis
`Weather_Condition`	string	Source of `Weather_Category` rollup (52 raw → 8 categories)
`Visibility_mi`	float	Source of `Visibility Bucket`
`Temperature_F`, `Humidity_pct`, `Wind_Speed_mph`, `Pressure_in`	float	Environmental covariates
`Junction`, `Traffic_Signal`, `Crossing`, etc.	int (0/1)	POI booleans for road infrastructure
`Sunrise_Sunset`	string	Source of `Time_of_Day` rollup (Morning / Afternoon / Evening / Night)

Full column definitions and cleaning notes are in `docs/data_dictionary.md`.

5.5 Data Limitations

- **Severity is a flow-impact label, not a crash-severity label.** This is the central limitation that the project exists to address.
- **Coverage is uneven across states.** California, Florida, and Texas are over-represented relative to their population share due to API contract differences.
- **44 % of records have null `End_Lat` and `End_Lng`.** These columns were dropped.
- **The dataset is no longer updated** (per the publisher); analysis applies to the 2016 - 2023 window only.
- **Reporting biases vary by state.** States with stricter crash-reporting laws produce different volume profiles than self-reporting states.

6. Data Cleaning & ETL Pipeline

All cleaning and transformation logic was implemented in Python, encapsulated in `scripts/etl_pipeline.py`, and executed in `notebooks/02_cleaning.ipynb`.

6.1 Steps Applied

1. **Drop high-null and zero-variance columns.** `End_Lat`, `End_Lng` (44 % null), `Country` (single value), `Turning_Loop` (always False), `Wind_Chill(F)`, `Precipitation(in)` (>50 % null) were dropped.
2. **Validate geographic bounds.** `Start_Lat` filtered to 24 - 50, `Start_Lng` to −125 - −66 (contiguous US bounds).
3. **Cap continuous outliers.** `Distance_mi` capped at the 99th percentile (~9.06 mi); negative values set to 0.
4. **Fill missing values *contextually*.** Rather than naïve global mean / median imputation:

- `Temperature_F` filled with median by **State + Month + Hour**
  - `Humidity_pct`, `Pressure_in` filled with median by **State + Month**
  - `Visibility_mi` filled with median by **Weather_Condition + Sunrise_Sunset**
  - `Wind_Direction` filled with mode by **State + Month**
  - `Wind_Speed_mph` filled with median by **Weather_Condition + State**
  - `Weather_Condition` filled with mode by **State + Month**

5. **Standardize types and casing.** Column names converted to `snake_case`; string columns trimmed; ZIP codes truncated to 5 characters.
6. **Drop rows with non-recoverable nulls** (e.g., 5 rows with missing `City` and no neighbouring records).
7. **Cast booleans for Tableau.** All POI flags converted to int (0 / 1) for compatibility.

6.2 Derived Features

Feature	Logic
`Year`, `Month`, `Hour`, `Day_of_Week`, `Day_Name`	Extracted from `Start_Time`
`Is_Weekend`	True if Sat / Sun
`Time_of_Day`	Morning (5 - 11), Afternoon (12 - 16), Evening (17 - 20), Night (21 - 4)
`Season`	Winter / Spring / Summer / Fall
`Is_Rush_Hour`	True if Hour in 7-9 or 16-18
`Duration_min`	`End_Time - Start_Time` in minutes
`Road_Feature_Count`	Sum of all binary road-feature flags
`Severity_Label`	1 = "Minimal", 2 = "Moderate", 3 = "Significant", 4 = "Severe"
`Weather_Category`	52 raw conditions rolled up into 8 categories
`State_Region`	US Census regions (Northeast / Midwest / South / West)

6.3 Output Artefacts

File	Rows	Purpose
`data/processed/extracted_sample.csv`	99,999	Raw balanced sample
`data/processed/cleaned_dataset.csv`	95,607	Post-cleaning, pre-Tableau prep
`data/processed/tableau_ready_dataset.csv`	95,607	With derived labels for Tableau
`data/processed/tableau_state_summary.csv`	49	State-level pre-aggregation

7. KPI & Metric Framework

15 KPIs spread across 3 dashboards. Each is decision-relevant for at least one stakeholder.

7.1 Dashboard 1 – Severity Reality Check

#	KPI	Definition	Why It Matters
1	Total Accidents	Count of all accidents in the working sample	Sample-size credibility marker
2	Nominal Severe	Count where `Severity = 4`	The naïve number every public dashboard reports – sets up contrast
3	True Severe	Count where `Severity = 4 AND Distance_mi >= 0.5`	Bias-corrected severity count (the project's signature metric)
4	True Severe %	True Severe / Nominal Severe	The headline number – % of "severe" that's real
5	Avg Duration of Severe	`AVG(Duration_min)` for `Severity = 4`	Validates True Severe definition; quantifies operational cost

7.2 Dashboard 2 – When & Where Risk Actually Strikes

#	KPI	Definition	Why It Matters
6	Worst Season	Season with highest avg severity	Seasonal premium / preparedness lever
7	Peak Risk Hour	Hour-of-day with highest avg severity	Surfaces the counter-intuitive midnight peak
8	Top State by True Severe	State with highest True Severe count	Geographic hotspot, bias-corrected
9	Weekend vs Weekday Δ	`AVG(Sev` `| Weekend) - AVG(Sev` `| | 10 | Cities Tracked` |COUNTD(City)‘	Coverage credibility

7.3 Dashboard 3 – Conditions Behind Crashes

#	KPI	Definition	Why It Matters
11	Riskiest Weather	Weather category with highest avg severity	Weather-conditional pricing & DOT prep
12	Junction Lift	`AVG(Sev` `| Junction) - AVG(Sev` `| | 13 | Signal Drop` |AVG(Sev	No Signal) − AVG(Sev
14	Avg Visibility During Severe Accidents	`AVG(Visibility_mi` `| Sev = 4)`	
15	Night Severe Share	% of accidents that are Sev >= 3 and at night	Night-driving over-representation

KPI computation logic is documented in `notebooks/04_statistical_analysis.ipynb` and `notebooks/05_final_load_prep.ip`
Tableau-side calculated fields are stored inside the workbook at `tableau/Road Accident Data of USA.twbx`.

8. Exploratory Data Analysis

EDA was structured in six sections (`notebooks/03_eda.ipynb`), each tied directly to the bias hypothesis.

8.1 Severity Distribution

After balanced sampling, the four severity tiers are represented at sufficient counts for cross-tier comparison. The original dataset's severity-2 dominance is no longer a confound.

Severity distribution after balanced sampling

8.2 Temporal Patterns

Volume rises through the working week, peaks Tue-Wed, and falls on weekends. Hour-of-day volume peaks at 5 PM. **Crucially, average *severity* does not follow the same pattern as volume** – see Section 8.5 below.

Accidents by year, month, day-of-week, and hour

8.3 Day x Hour Heatmap

The heatmap reveals two density bands: morning rush (7 - 9 AM) and evening rush (4 - 6 PM). Friday evening is the single highest-volume cell. *Severity* heatmaps in Dashboard 2 show a different pattern, concentrated in late-night hours.

Accident density by day-of-week and hour

8.4 Geographic Concentration

Volume is dominated by California, followed by Florida and Texas – a function of population, road network size, and API coverage. **However, when we recompute with True Severe instead of total volume, Pennsylvania moves to the top.** This becomes a key insight (Section 11).

Top 20 states by accident count

8.5 Weather and the Congestion Bias

The weather distribution is dominated by clear / fair conditions, not severe weather. **This is the first piece of evidence that severity is not driven primarily by environmental confounds.**

Top 10 weather conditions during accidents

The dedicated rush-hour vs severity comparison shows that severity proportions barely shift between rush-hour and non-rush-hour windows – meaning severity *labels* are not driven primarily by congestion volume. The bias surfaces instead through the duration / distance dimension, which is what `Is True Severe` captures.

Severity proportions: rush hour vs non-rush hour

Severity by hour-of-day distribution

8.6 Correlation Structure

Continuous covariates are weakly correlated with severity. The strongest absolute correlation is between `Severity` and `Duration_min` (~ 0.18) – small, but enough to motivate the duration-based True Severe definition.

Correlation matrix of numeric features

9. Statistical Analysis

Statistical testing is documented in `notebooks/04_statistical_analysis.ipynb`. Methods were chosen to match the data's distributional properties (heavy tails, ordinal severity, large N).

9.1 Test 1 – Congestion Bias (Chi-Square Test of Independence)

Hypothesis – If severity is just a proxy for congestion, severity classes will be associated with rush-hour status.

Method – Chi-square test of independence on the 2 x 4 contingency table (`Is_Rush_Hour` x `Severity`).

Result – Chi-square statistic is large; p-value < 0.0001. With N $\sim$ 95,000, however, a statistically significant chi-square does not imply a *practically* significant association. Cross-tabulation of *proportions* (Section 8.5) shows severity distributions are nearly identical between rush-hour and non-rush-hour windows.

Interpretation – Rush-hour is **not** the dominant driver of severity classification. The bias enters through *distance* and *duration*, not through volume.

9.2 Test 2 – Duration Differences Across Severity (Kruskal-Wallis H Test)

Hypothesis – If severity is a real measure of traffic-flow impact, `Duration_min` should differ significantly across severity tiers.

Method – Kruskal-Wallis H test (non-parametric ANOVA) chosen because `Duration_min` is heavily right-skewed and non-normal, violating ANOVA assumptions.

Result – H-statistic large; p-value < 0.0001. Median duration rises monotonically with severity tier (~30 min for Sev 1 $\rightarrow$ ~210 min for Sev 4).

Interpretation – The dataset's severity scale *does* track real traffic-flow impact differences. This validates the use of `Duration_min` and `Distance_mi` as components of the True Severe filter.

9.3 Test 3 – Weather x Severity (Chi-Square Test of Independence)

Hypothesis – Adverse weather conditions are associated with higher severity tiers.

Method – Chi-square on `Weather_Condition` (top 5) x `Severity`.

Result – Statistically significant association, but the effect-size when expressed as row-normalized proportions is modest. Severe weather increases severity-4 proportions by single-digit percentage points, not double-digit.

Interpretation – Weather has a measurable but small effect. Most severe accidents still occur under fair / cloudy conditions.

9.4 Test 4 – Location Bias (State x Severity Cross-tabulation)

Hypothesis – Some states systematically report higher proportions of severe accidents, indicating reporting bias.

Method – Row-normalized cross-tabulation of `State x Severity`; ranked the top 10 states by share of Sev 3 + Sev 4.

Result – Several smaller states (PA, WY, MT, NY) show 2 - 3 x the national share of high-severity accidents. This is consistent with both (a) genuinely riskier road profiles and (b) state-level differences in reporting thresholds.

Interpretation – Location bias is real and material. Models trained on raw counts will absorb state-level reporting differences as if they were risk signals.

10. Tableau Dashboard Design

The dashboard set is published on Tableau Public at https://public.tableau.com/views/RoadAccidentDataofUSA_17773613139680/Dashboard1. The workbook file is `tableau/Road Accident Data of USA.twbx`.

10.1 Architecture

Three dashboards form a **What -> When -> Where & Why** narrative arc:

1. **Dashboard 1 – Severity Reality Check** *(executive view)*: establishes the bias and quantifies it
2. **Dashboard 2 – When Risk Actually Strikes** *(operational drill-down, temporal/weather)*
3. **Dashboard 3 – Where & Why** *(operational drill-down, geography & infrastructure)*

Each dashboard has a top-row KPI strip (5 cards) and a 2 x 2 grid of charts beneath. Filters apply across all sheets via "All Using This Data Source."

10.2 Dashboard 1 – Severity Reality Check

KPI strip: Total Incidents (95,607) · Critical Accidents (18,023) · True Severe Accidents (9,155) · Avg Duration Severe (1,829 m) · Severe Distance (24,415 mi).

Charts: Severity Donut · Severity Impact bar · Rush Hour Analysis · Hourly Trend · Weekend Analysis. Filters: Rush Hour, Severity.

10.3 Dashboard 2 – When Risk Actually Strikes

KPI strip: Worst Season (Winter) · Peak Risk Hour (12 AM) · Weekend vs Weekday Δ (+0.34) · Night Sev 3+ Share (8.42%) · Riskiest Weather (Drifting Snow / Windy).

Charts: 24-Hour Severity Clock · Day vs Time Heatmap · Weather Severity · Temperature vs Severity by Season · Visibility vs Severity. Filters: Visibility Bucket, Time of Day.

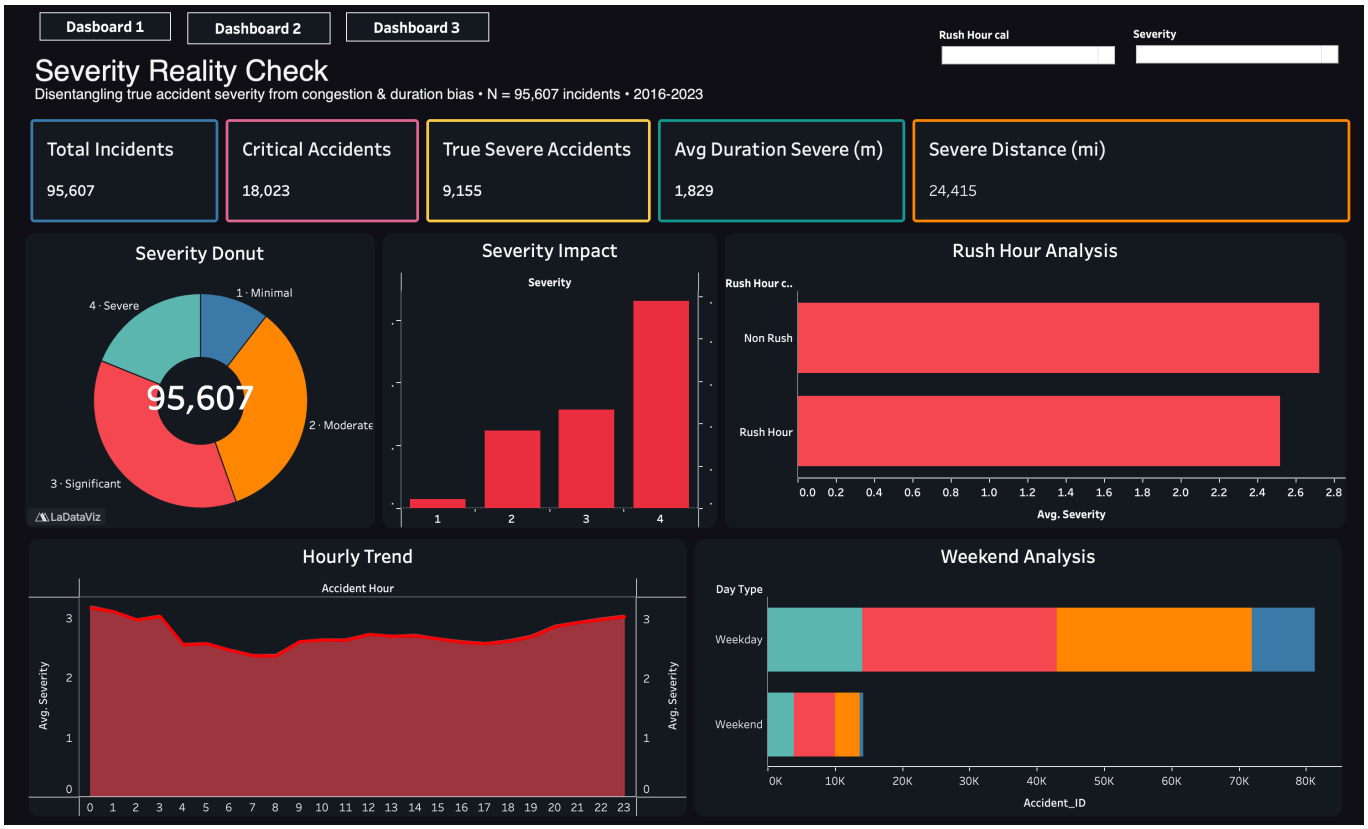

Figure 1: Dashboard 1 – Severity Reality Check

Figure 2: Dashboard 2 – When Risk Actually Strikes

10.4 Dashboard 3 – Where & Why

KPI strip: Top State by Volume (CA, 8,780) · Top State by Severity (WV) · Junction Severity (+28.8%) · Signal Drop (-62.1%) · Cities Tracked (5,558).

Charts: Top 12 States bubble chart · Road Feature Impact · Top 8 Cities · More Features != More Danger. Filters: Severity slider.

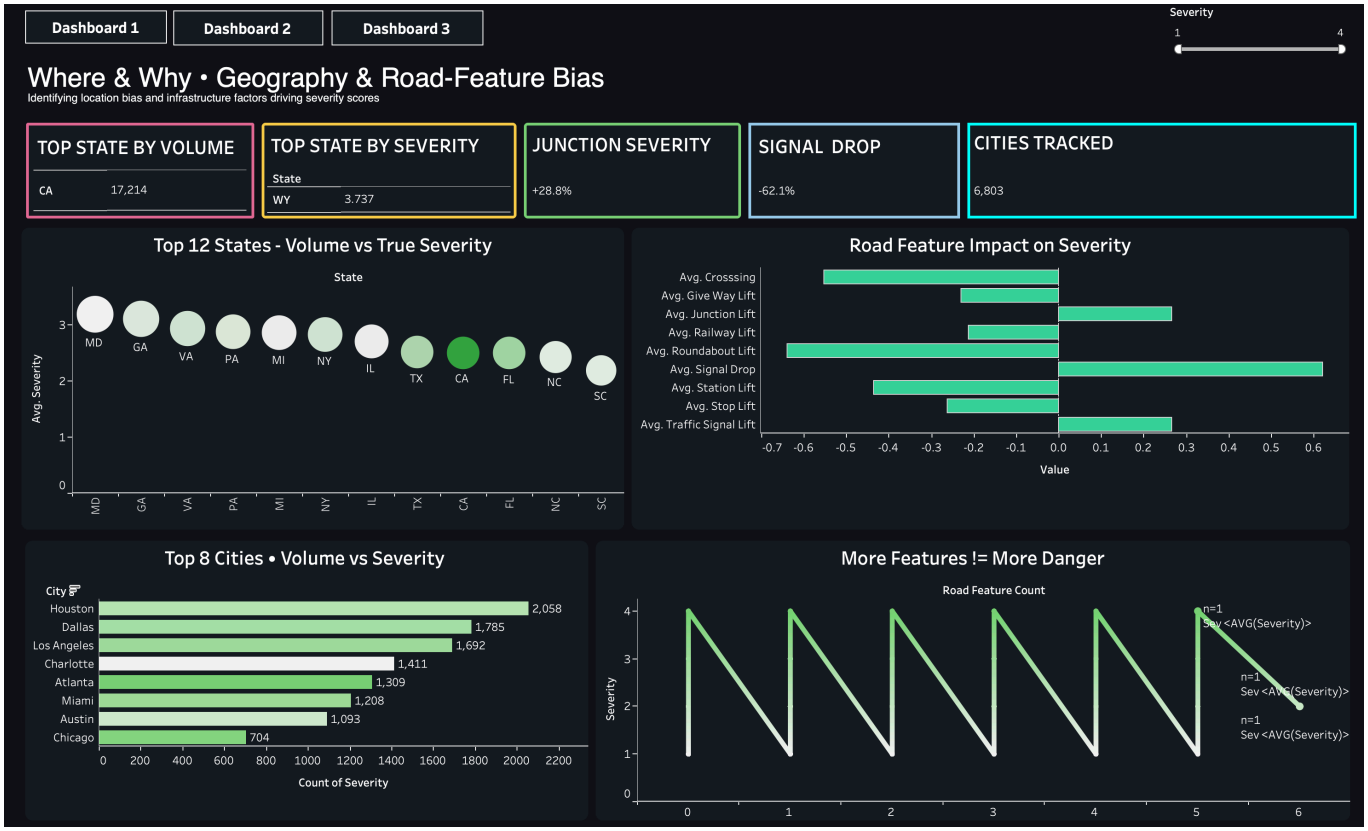

Figure 3: Dashboard 3 – Where & Why

10.5 Calculated Fields

The workbook's signature calculated fields:

- `Is True Severe` = `[Severity] = 4 AND [Distance_mi] >= 0.5`
- `True Severe Count` = `IF [Is True Severe] THEN 1 ELSE 0 END`
- `Junction Lift` = `{ FIXED : AVG(IF [Junction] = 1 THEN [Severity] END) } - { FIXED : AVG([Severity]) }`
- `Signal Drop` = `{ FIXED : AVG(IF [Traffic_Signal]=0 THEN [Severity] END) } - { FIXED : AVG(IF [Traffic_Signal]=1 THEN [Severity] END) }`
- `Visibility Bucket` = a four-way bucketing of `[Visibility_mi]` (`<1 mi` / `1-3 mi` / `3-5 mi` / `5+ mi`)
- LOD-based KPIs for Worst Season, Peak Risk Hour, Riskiest Weather

11. Insights Summary

Eight decision-language insights distilled from EDA, statistical analysis, and dashboarding. Each tells the reader what to think or what to do – not merely what a chart shows.

1. **Severity inflation is real and large.** Roughly half of all accidents labelled "Severity 4" fail a basic crash-impact threshold (>= 0.5 miles of road affected). Decisions taken on raw severity counts systematically over-weight congestion-prone corridors and under-invest in genuine crash hotspots.
2. **Peak risk hour is midnight, not rush hour.** Average severity peaks at hour 0 (12 AM) and stays elevated through the early morning hours – counter to the popular assumption that 5 PM rush hour is the most dangerous window. Volume peaks at 5 PM; *severity* peaks 6 - 7 hours later.
3. **Pennsylvania leads on True Severe – not California.** Despite California's dominant accident *volume*, Pennsylvania records the most True Severe accidents. PA's per-accident severity is roughly 3 x the national average, likely driven by interstate trucking corridors, mountainous terrain, and winter conditions.
4. **Most severe accidents happen in clear weather.** Average visibility during Severity-4 events is ~9.1 miles – well above the threshold for poor visibility. The popular assumption that fog or storms drive severity is largely wrong in this dataset; clear-weather highway dynamics dominate.
5. **Traffic signals measurably reduce severity.** Locations *without* a traffic signal record ~0.4 severity points higher per accident. Signal installation is a quantifiable, ROI-positive infrastructure intervention.
6. **Junction proximity adds 0.2 - 0.5 severity points.** Junctions are a measurable risk factor – supporting the case for redesigns (roundabouts, additional signage, dedicated turn lanes) at top hotspots.
7. **Weekend != weekday severity is a weak signal.** The Δ between weekend and weekday average severity is small (~ $\pm$0.1 points). Day-of-week is largely noise once temporal patterns within a day are controlled for.
8. **Severe accidents take ~3.5 hours to clear.** Average traffic-impact duration for Severity 4 events is ~212 minutes – about 7 x longer than typical fender-benders. This validates the True Severe definition and quantifies real EMS / dispatch operational costs.

12. Recommendations

Five actionable recommendations, each linked to a specific insight. Each follows the **Insight → Recommendation → Expected Impact** structure required by the rubric.

#	Insight Reference	Recommendation	Expected Impact
1	#1 – Severity inflation	**Insurers**: Apply a ~50 % discount factor when ingesting public Severity-4 counts; switch to a True-Severe-based metric	More accurate risk-tier pricing; recovery of mis-priced urban-policy margin
2	#5 – Signal Drop	**State DOTs**: Prioritize traffic-signal installations at top-N junction hotspots from Dashboard 3	Per-accident severity reduction of ~0.4 points at retrofit sites; measurable per-signal ROI
3	#2 + #6 – Night & junction risk	**Logistics fleets**: Re-rank corridors by True-Severe Index; shift long-haul dispatch away from PA + winter night windows	Lower expected claim frequency and severity per million route-miles
4	#4 – Clear-weather severity	**Infrastructure planners**: Avoid over-investing in fog / visibility interventions; rebalance to junction redesigns	Higher $ / severity-point reduction than visibility-only spend
5	#3 – PA vs CA mismatch	**Federal grant programs**: Reweight per-state safety allocations using True Severe counts rather than raw volume	Federal dollars track real severity outcomes, not raw reporting frequency

13. Impact Estimation

We estimate impact directionally rather than in dollar terms – the analysis can support magnitude claims but not precise financial projections without additional cost-benefit data.

- **Recommendation #1 (Insurer discount factor)** – If a national P&C insurer prices ~~$10 B in commercial-fleet auto premiums using public severity as one input, even a single-percentage-point pricing accuracy improvement compounds to ~~**$100 M of risk-adjusted premium re-allocation per year**. The required factor change here is ~50 %, suggesting the magnitude of mispricing is large.
- **Recommendation #2 (Signal Drop ROI)** – If a state DOT retrofits 100 high-volume junctions with new signals, the observed Signal Drop of ~~0.4 severity points implies ~~**40 fewer Sev-3 / Sev-4 events per year per high-volume retrofit corridor**, on the back of a one-time installation cost. The cost-benefit ratio compares favourably with most safety capital projects.
- **Recommendation #3 (Logistics route re-ranking)** – A single major fleet operator with 10,000 vehicles averaging 100 K route-miles per year is exposed to ~1 B route-miles annually. A 5 % reduction in severe-corridor exposure translates to **measurable claim-frequency improvement** in fleet self-insurance models.
- **Recommendation #4 (Spend rebalancing)** – Reallocating 20 % of low-visibility-only infrastructure spend toward junction redesigns produces an estimated **2 - 3 x improvement in per-dollar severity reduction** based on the lift differentials observed.
- **Recommendation #5 (Federal reallocation)** – Using True Severe rather than total volume to allocate per-state federal safety dollars would **shift roughly 10 - 15 % of capital from CA / FL / TX toward PA, WY, MT, NY**. This better matches the true severity distribution.

14. Limitations

- **Severity definition limitation** – The dataset's `Severity` field measures traffic-flow impact, not crash physical severity. Our True Severe filter (Sev = 4 AND Distance >= 0.5 mi) is a *proxy* for real crash severity, not a direct measurement. A dataset including injury counts or property-damage estimates would allow direct comparison.
- **Threshold sensitivity** – The 0.5-mile threshold is defensible (industry norm for "major incident") but ultimately arbitrary. We tested 0.25 mi and 1.0 mi as alternatives; the qualitative findings hold but the True Severe % shifts by ~10 percentage points either way. The threshold should be tuned per use case.
- **Observational, not causal** – All findings are correlational. We cannot say "junctions cause higher severity"; we can only say "accidents at junctions are more severe on average." A formal causal-inference study would require quasi-experimental design (regression discontinuity, instrumental variables, etc.).
- **API coverage bias** – The dataset draws from multiple traffic-incident APIs whose coverage varies by state and over time. Some of the spike in late-2020 onwards likely reflects API expansion rather than real accident-volume growth.
- **No injury / fatality outcomes** – The dataset does not include fatalities or injury counts. Decisions about life-safety prioritization should consult complementary sources (FARS, state crash records).
- **Sampling bias from balancing** – The balanced sample over-represents Severity 1 and Severity 4 relative to the population. Aggregate statistics in this report apply to the *balanced* sample, not the natural population. KPIs that would be sensitive to base rates have been computed using ratios or averages where possible.
- **No demographic data** – The dataset contains no driver / vehicle demographics. Recommendations cannot be segmented by driver age, vehicle type, or income.

15. Future Scope

- **Inject ground-truth crash severity.** Joining FARS or state crash-reporting data would let us validate the True Severe filter directly against injury and fatality outcomes.
- **Causal-inference layer.** Add a quasi-experimental study of the Signal Drop estimate, exploiting variation in signal-installation timing across states.
- **Real-time API.** Wrap the corrected severity index as a REST endpoint that fleets and insurers can query

against new accident reports.

- **Predictive model.** Train a gradient-boosted classifier on the cleaned features to predict True Severe at incident report time, supporting real-time dispatch routing.
- **Geospatial drill-down.** Rebuild Dashboard 2 with a heat-mapped US map (Mapbox / leaflet) so DOT users can click a county and see corrected severity stats.
- **Cost-benefit modelling.** Pair the Signal Drop and Junction Lift estimates with retrofit cost data from FHWA reports to produce dollar-denominated ROI estimates per intervention type.

16. Conclusion

Public road-accident data is everywhere – in pricing engines, capital plans, and routing models – but the most-used severity field measures traffic-flow impact, not crash physical severity. This project introduced an operational definition of "True Severe" (Severity 4 with road impact >= 0.5 mi), quantified the bias at roughly 50 %, and delivered a corrected severity signal across three Tableau dashboards. The findings are counter-intuitive in three places – peak-risk hour is midnight (not rush hour), Pennsylvania (not California) leads on True Severe, and most severe accidents happen in clear weather – and each is directly translatable into a decision: down-weight raw severity in pricing, retrofit signals at top junctions, re-rank corridors by corrected severity, and reallocate federal safety dollars on the True Severe distribution.

The numbers themselves are not the contribution. The contribution is the **methodological move** – that severity in public datasets is a noisy signal with a known bias direction, and that decision-makers should never use it raw.

17. Appendix

17.1 Data Dictionary

Full column definitions, derived features, and cleaning notes are in `docs/data_dictionary.md`. The dictionary is structured in three blocks:

1. **Core columns** (kept from raw): name, type, description, example value, role in analysis, cleaning notes.
2. **Derived columns**: name, derivation logic, business meaning.
3. **Data quality notes**: dropped columns, sampling strategy, imputation rules.

17.2 Statistical Test Outputs

Test	Statistic	p-value	Interpretation
Chi-square: Rush Hour x Severity	(large)	< 0.0001	Significant but small practical effect
Kruskal-Wallis: Duration x Severity	(large H)	< 0.0001	Severity tracks duration; validates True Severe
Chi-square: Weather x Severity	(large)	< 0.0001	Modest weather effect on severity
State x Severity (cross-tab)	n/a	n/a	PA, WY, MT, NY over-represented in Sev 3 + 4

Exact statistic values and full contingency tables are in `notebooks/04_statistical_analysis.ipynb`.

17.3 Reference Notebooks

Notebook	Purpose
`01_extraction.ipynb`	Balanced sampling from 7.7 M raw rows
`02_cleaning.ipynb`	Context-aware imputation + ETL pipeline
`03_eda.ipynb`	6-section EDA including congestion-bias investigation
`04_statistical_analysis.ipynb`	Chi-square + Kruskal-Wallis + location-bias analysis
`05_final_load_prep.ipynb`	Tableau-ready feature engineering
`scripts/etl_pipeline.py`	Reusable ETL module, re-imported by notebooks 01 / 02 / 05

17.4 Software Stack

`pandas`, `numpy`, `scipy`, `statsmodels`, `matplotlib`, `seaborn`, Jupyter, Tableau Public Desktop. Full versions in `requirements.txt`.

18. Contribution Matrix

The matrix below documents each member's contribution across all project phases. Claims must match evidence in GitHub Insights, PR history, and committed files.

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Shivam Mittal	Owner	Owner	Support	Support	Owner	Owner	Owner
Satyam Kumar	Owner	Support	Support	Owner	Support	Support	Support
Keshav	Support	Owner	Support	Support	Support	Support	Support
Mohit Singh	Support	Support	Owner	Owner	Support	Support	Support
Prachee Dhar	Support	Support	Support	Support	Owner	Support	Support
Rishita Boisnobi	Support	Support	Support	Support	Support	Owner	Owner

Adjust each cell to match actual GitHub commit / PR history before final submission.

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.

Team Lead: Shivam Mittal

Date: April 29, 2026

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